

1 Edit Operations

Overview

HYDRA menu	Order management → Order management → Edit operations
FEDRA menu	Detailed Scheduling → Order management → Edit operations
Transaction code	edop
Function authorization	edop

Purpose

The term process, work step or operation describes a workflow that is designed to perform a task in a work system. During this workflow one quantity unit of an order is produced.

You use this function to add new operations to an order or to edit data in existing operations.

Integration

Operations are planned in planning functions and optionally posted on shop floor terminals; their purpose is to facilitate status tracking and to record quantities and activities, which are usually uploaded to higher-level systems.

Requirements

The following requirements must be met when adding a new operation:

- The higher-level order must already have been created.
- If you use order sequences (project-specific; depending on the license), the sequence of the operation must already exist.
- The workplace/machine or group where you want to plan the operation has already been created in the system.

The authorization for the responsibility area is assigned to you and you are authorized to display the data.

Selection criteria

The application provides the following selection criteria:

Order

Enter the order number of the order that includes the operations you want to display. You can also use wildcards.

Operation

You can optionally enter the operation number of the operation that you want to display or edit. You can also use wildcards.

Sequence

If your system is set up to use sequences (depending on the license), you can enter the sequence number here. The system then selects the operations assigned to the sequence number entered. If your system is not set up for sequences, leave this field empty.

Show split OPs

If you use the function to split operations (requires license), you can use this option to define whether you want to display the split-master-operations only or also the included split operations.

Checking the responsibility area

During the selection, the responsibility area defined for the operation is checked.

Field descriptions

The fields of the operation are described [here](#).



Only selected data is available in the table:

- Order
- Sequence
- Operation
- Split
- Processing code
- Locked
- Fixed
- Group
- Workplace
- Control

Editing functions

To create or edit operations, use the buttons provided.

If a responsibility area is stored for the order, the editing of data is only possible if the options to display, insert, modify and delete are enabled in the configuration of the responsibility areas or profiles.

Toolbar



Edit orders

Function authorization: edor.*

Calls the application [Edit orders](#) for the selected order.



Edit order sequences

Function authorization: edseq.*

Calls the application [Edit order sequences](#) for the selected order.



Edit long texts of operations

Function authorization: edoptx.*

Calls the application [Edit long texts of operations](#).



Edit components

Function authorization: edopcomp.*

Calls the application [Edit components](#).



Edit production resources and tools

Function authorization: edopres.*

Calls the application [Edit production resources and tools](#).



Order information

Function authorization: orin

Calls the application [Order information](#) for the selected order.



Change operation status

Function authorization: op.statchg

Function to [change the operation status](#).

**Lock**

Function authorization: op.lock

Use the button [Lock operation](#) to lock one or several selected operations.

**Unlock**

Function authorization: op.unlock

The button [Unlock operation](#) unlocks one or several selected operations.

**Split operation**

Function authorization: op.split

Calls the function to split the operation. For further information, refer to the relevant [documentation](#).

**Dissolve split OP**

Function authorization: op.splitdissolve

Undoes the operation split. For further information, refer to the relevant [documentation](#).

Adding an operation

Transferring order header data

The following data is transferred from the order header in the operation when a new operation is created:

- Order type
- Base quantity unit
- Article if the article number is not explicitly defined for the operation.
- Article designation if the article designation is not explicitly defined for the operation.
- Material type if it is not explicitly defined for the operation.
- Customer name
- Priority, if the priority control is set to *order-related* for the order type..



Any priority that may have been entered will be ignored!

Transferring default data

Default data is taken from a template or from the processing code, if one exists, and transferred to the operation when an operation is created. The data is transferred in the following order:

- Values are transferred from the template (if available).

If you add a new operation (manually or via interface), all values are transferred that can be edited in the template and that are not entered manually (explicitly) or transferred via interface .

- Values are transferred from the *processing code* (if one exists); doing so will overwrite any values set in the template. The following values are transferred from the processing code to the operation:
 - Underdelivery
 - Reaction to underdelivery
 - Overdelivery
 - Reaction to overdelivery
 - External processing
 - Recordable
 - Can be logged on several times
 - Can be split
 - Serial number obligation
 - Batch management requirement
 - Target quantity update*
 - Sequencing list* is no longer evaluated for display in the sequencing list; instead, the system directly accesses the separate configuration tables.

Note: Values marked with * are not displayed for the operation and therefore they also cannot be changed.

- Transfer of the values transmitted explicitly (either entered manually or transmitted via PPS interface); any values that were previously transferred from the template or the processing code will be ignored and overwritten.

Target quantity comparison

If the target quantity comparison is enabled for the preceding operation, then any target quantity that is entered is ignored and instead the target quantity of the preceding OP is used when you add an operation.

Identifying the transport time

To identify the transport time between two operations, you can store a transport matrix . This is part of a HYDRA customization. This transport time is then integrated during lead time scheduling.

When a new order or operation is created, the transport time is calculated using this matrix and then transferred to the operation. If you change the transport matrix later on, this will have no effect on already existing operations.

If a transport time is configured greater than zero in the ERP system and if operations are then transferred, these transport times are transferred to the database. Otherwise, the time is calculated using the transport matrix.

You need not use master data, you can also explicitly change the values for the operation. Note: any values changed explicitly are overwritten when you re-plan an operation to another machine group.

Setting the planned start data (used in the [sequencing list on the terminal](#))

When a new data record is added, the system tries to identify planned start dates and to use them as default values for the sorting of the sequencing list. This process is based on the following logic:

- It is checked whether or not the planned start date and the planned start time are empty.
 - If yes:
 - The earliest start date (date) is used as planned start date.
 - The earliest start date (time) is used as planned start time.
- It is checked whether or not the planned end date and the planned end time are empty:
 - If yes:
 - The latest end date (date) is used as planned end date.
 - The latest end date (time) is used as planned end time.
- For the sorting of the sequencing list, the planned start date and the planned start time are entered in separate fields that cannot be changed.

The used date fields, the corresponding BAPI acronyms and database fields can be found in the document dealing with the technical [background information on the sequencing list](#).

Editing an operation

If you edit an operation, the default data (if available) is taken from the template or from the processing code and transferred to the operation in the following order.

- If you change the planned group or if you change the workplace and the new workplace is included in a different group, then the system transfers the following values (if available) from the template:
 - Waiting time formula
 - Setup time formula
 - Processing time formula
 - Inspection time formula (only for information purposes).
 - Teardown time formula
 - Target cycle formula
 - Formula for the [remaining run time](#)
 - Formula for the second remaining run time
 - Max. synchronization time
 - Default value key
- If the planned group was changed, the system will also update the value plan_werk (internally managed in the order backlog) in which the company for the modified [Group](#) is identified and transferred.
- If the processing code was changed for the operation, the values of the processing code are transferred (see above).
- The final step involves the transfer of the values transmitted explicitly (either entered manually or transmitted via ERP interface); this will ignore/overwrite any values that were previously transferred from the template or the processing code.

If the group of an operation is changed, the transport time is recalculated. The transport time stored in the transport matrix is then used. Any transport time defined for the preceding OP will be ignored/discarded.



If you change the target quantity (P), the target quantity is only converted automatically to the other quantity units if the fields have been emptied manually beforehand.

Transferring order header data

If you change an operation, only the values below are transferred from the order header to the operations:

- Priority, if the priority control is set to *order-related* for the order type..



Any priority that may have been entered will be ignored!

- Customer name

The base quantity unit is not changed, because it is very unlikely that a change of this kind would ever happen in reality. The material type is not modified, because in MPL it may vary from one OP to the next OP.

General checks run when an operation is saved

Checking the existence of workplace or group

If a workplace was entered, then the system checks whether the workplace exists. If yes, the workplace group of the workplace is transferred to the operation. In any other case, the saving process is interrupted with an error message.

If no workplace was defined, but instead only a group, then the system checks the validity of the group that was entered. This means: it checks if the group exists in the system. If no, the change is rejected and an error message is issued.

Checking priority management

If priority management was activated for the order type as part of the customizing process using the identifier with the same name ADE_AUFTRAGSARTEN.PRIO_STEUERUNG[2,2] and if the priority control was configured as order-related PRIO_STEUERUNG[1,1] = 'U' , then the system checks whether the defined priority is permitted when a new order is created manually or an order is changed. If the maximum number is violated, the action will be rejected.

If the order is transferred from the ERP interface and the maximum number is exceeded, the order is not be refused as a result of this validation check. In this case, however, the priority is automatically set to 1.

Ability to modify an operation

By default, the following operations cannot be modified:

- OPs that are currently logged on (status with control indicator L) and
- OPs that are automatically interrupted (status with control indicator F).

Checking the formula values transferred

As of b_anr.dll version 8.1.1.359.

If you add or change an operation, the system checks the values passed in the formula fields (if specified). The system checks if the values are available in the [formula management](#). The validation check is performed for the following formula fields:

- Setup time formula (BAPI identification ANR.RUEZ:EXPR)

- Processing time formula (BAPI identification ANR.BEARBZ:EXPR)
- Inspection time formula (BAPI identification ANR.PZ:EXPR)
- Teardown time formula (BAPI identification ANR.ABRZ:EXPR)
- RRT1 formula (BAPI identification ANR.RLZ:EXPR)
- RRT2 formula (BAPI identification ANR.RLZ2:EXPR)
- Waiting time formula (BAPI identification ANR.WARTZ:EXPR)
- Target cycle formula (BAPI identification ANR.SZY:EXPR)

In addition to the formula fields, the following BAPI identifications are also checked:

- ANR.TE.EXPR
- ANR.TR.EXPR
- ANR.TEB.EXPR
- ANR.TRB.EXPR

If at least one of the values entered is not available in the formula management, the request is rejected and an error message is issued (return code 901).

You can deactivate the validation check using the following entry in the INI configuration:

Parameter name	Value
INI name	BDE
Section	ANR_BAPI
Key	CHECK_FORMULAS
Value	N → Nein / No (check is disabled)
Active	Yes
Comment	(optional)

Setting the planned start data (used in the [sequencing list on the terminal](#))

When an existing data record is changed, the system tries to identify planned start dates, to update them and to use them as default values for the sorting of the sequencing list. The basic logic depends on the planning function configuration in the master record of the operation's workplace:

- Planning function "N" – no planning
 - It is checked whether or not the planned start date and the planned start time are empty.
 - If yes:

- The earliest start date (date) is used as planned start date.
 - The earliest start date (time) is used as planned start time.
- It is checked whether or not the planned end date and the planned end time are empty:
 - If yes:
 - The latest end date (date) is used as planned end date.
 - The latest end date (time) is used as planned end time.
 - For the sorting of the sequencing list, the planned start date and the planned start time are entered in separate fields that cannot be changed.
- Planning function "P" / "H" / "T" / "A":
 - If the planned start date was changed, the planned start date and the planned start time are entered in separate fields that cannot be changed.



If no workplace is defined for the operation, an identical processing is performed as with planning function "N".

The used date fields, the corresponding BAPI acronyms and database fields can be found in the document dealing with the technical [background information on the sequencing list](#).

Deleting operations

When an operation is deleted, the following points must be considered:

- By default, you can only delete an operation if the operation is not logged on, i.e. the operation is not in status "running" and not in status "automatically interrupted".
- You cannot delete a split operation. To delete a split operation, you must dissolve the operation using the relevant split functionality.
- If a split master is deleted, the split OPs are also deleted.
- You cannot delete a merged operation. You must dissolve it.
- If you manually delete the last operation of an order on the client, the order header is not deleted. It must be deleted explicitly.

By default, deleting an operation means that an item is physically deleted from the database. The following data is deleted:

- Backlog of orders
- Order status
- Assigned material components

- Assigned production resources and tools
- Assigned long texts
- Resource allocation for this operation in the shop floor scheduling (HLS)

The log data (Tabelle ade_protokoll) is not automatically deleted if an operation is deleted. The log data is transferred to the long-term table or deleted from the database in the course of the cyclic archiving/deletion runs.

Deleting orders via delete action "D" if last OP is deleted

As of b_anr.dll version 8.1.1.358.

If you "delete" the order header via MLE interface (PPS=J), only physically deleted operations are integrated. For operations that were only deleted logically (using delete action=D), no processing is available.

Use the following entry in the INI configuration to integrate also operations that were only logically deleted (delete action=D):

Parameter name	Value
INI name	BDE
Section	ANR_BAPI
Key	ORDER_BAPI_DELETE_WITH_DELETE_ACTION_D
Value	J → Ja / Yes:
Active	Yes
Comment	Delete Order when OP in status D

HYDRA: as of service pack >13

FEDRA: as of version 1.1

This processing is activated by default for new systems. You can deactivate this processing, if required. The processing is not automatically activated with subsequent updates.